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About Nexbash

Nexbash Systems is an enterprise technology partner that turns complex data into operational advantage by delivering enterprise-grade Artificial Intelligence, Geospatial, and Software engineering services to the organizations that manage critical infrastructure, public services, and large-scale operations.. Founded in 2019 and headquartered in Dubai, we help public and private sector clients convert complex spatial and operational data into reliable, production-ready systems that drive measurable outcomes.

We combine our technical depth of AI & ML, Data science, Geo-Informatics Systems, and cloud-native engineering, etc. with domain knowledge by working alongside operators, planners, and clinicians to embed intelligence into everyday workflows. Nexbash partners with leading cloud and GIS vendors to ensure enterprise compatibility, scalability, and compliance. Backed by experienced engineers, data scientists, and system designers, Nexbash helps organizations reduce risk, accelerate decisions, and turn data investments into sustained operational advantage.



OPERATIONAL IMPACT

We deliver platforms that move KPIs, not just prototypes.



INTEGRATED EXPERTISE

AI and spatial engineering delivered with production MLOps and secure SDLC.



ENTERPRISE DISCIPLINE

Governance, SLAs, and compliance built into every engagement.



HUMAN-CENTERED DESIGN

Solutions engineered for adoption by real users in real environments.

Welcome Message

Canva



Mr. Ahmed Raza

CEO Nexbash Systems

“At Nexbash Systems, we believe technology should change the way people see the world. It’s not just about building tools, it’s about unlocking possibilities. Our vision is to bring AI and geospatial intelligence together in a way that feels intuitive, powerful, and transformative, helping enterprises solve problems they didn’t even know they could solve.

We focus on creating production-grade solutions that are elegant in design, scalable by nature, and impactful by result. Because when technology is built with purpose and clarity, it doesn’t just keep up with the future ... it helps shape it.”



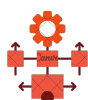
Vision

To become a trusted partner for organizations that rely on complex spatial and operational data, enabling them to use AI-driven and geospatial intelligence confidently, consistently, and at scale. We envision a future where data is not an experimental asset, but a dependable part of daily operations for protecting infrastructure, serving communities, and managing critical resources.

Mission

To help organizations make reliable use of AI and geospatial intelligence in their daily operations by turning complex data into systems that are trusted, adopted, and impactful. Our mission is centered on enabling long-term operational value, not one-off deployments, by ensuring intelligence is usable, measurable, and sustainable across the enterprise.

01



Build for real environments

Our mission is to support how organizations actually function within regulatory, technical, and operational constraints.

02



Value that can be measured

We focus on outcomes that matter to the business, with intelligence tied directly to performance, efficiency, and decision-making.

03



Create Impact that lasts

We seek to create systems that continue to deliver value over time through stability, knowledge continuity, and long-term support.

04



From pilots to operations

We aim to move AI and geospatial capabilities out of pilot phases and into dependable, production-ready use.



Core Capabilities

Nexbash Systems offers a comprehensive suite of services designed to transform complex data into actionable intelligence. Our integrated approach ensures seamless execution from concept to deployment, driving tangible results for our clients.

AI & MACHINE LEARNING

Design and development of custom machine learning solutions to support automation, decision-making, and predictive insights, including computer vision, NLP, and analytics pipelines.

GEOSPATIAL SYSTEMS

Development of advanced geospatial platforms to support mapping, spatial analysis, and location-based intelligence, using both Esri and open-source technologies.

DATA SCIENCE & ANALYTICS

Building data pipelines and analytics systems that help businesses understand patterns, monitor performance, and make data-driven decisions through dashboards and reporting tools.

BLOCKCHAIN & CRYPTO

Implementation of blockchain-based solutions for secure record-keeping, traceability, and digital asset experimentation, including smart contracts and tokenization prototypes.

WEB & MOBILE ENGINEERING

Enterprise-grade web and mobile applications designed to support day-to-day business operations, including responsive web platforms, native iOS/Android apps, and cross-platform solutions.

DESIGN UX/UI

User-focused design for enterprise systems, covering research, interface design, and usability testing to ensure products are intuitive, accessible, and easy to use.

QUALITY ASSURANCE

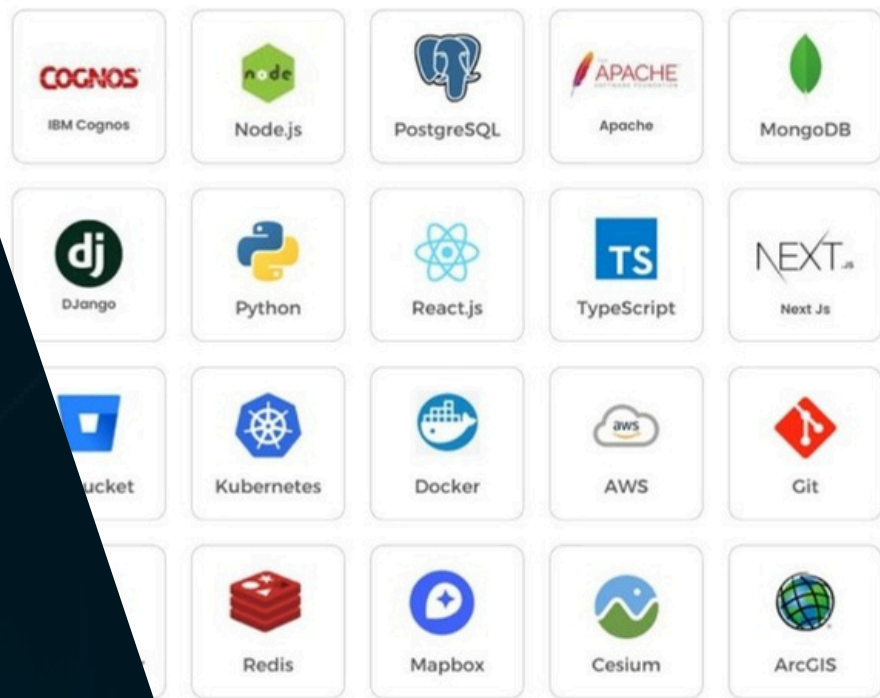
Comprehensive testing services to ensure software reliability, performance, and security, integrated seamlessly into development and release workflows.

DEVOPS & CLOUD

Cloud-native infrastructure and DevOps practices to support reliable deployments, system monitoring, scalability, and cost-efficient operations.

Our Tech Stack

We develop using modern, modular technologies – fully powered by our proprietary and partner-led solutions.



Cloud & Infrastructure

AWS · Microsoft Azure · Google Cloud · Kubernetes · Docker · Terraform

Data & ML

Spark · Kafka · TensorFlow · PyTorch · MLflow · Feature stores

Geospatial

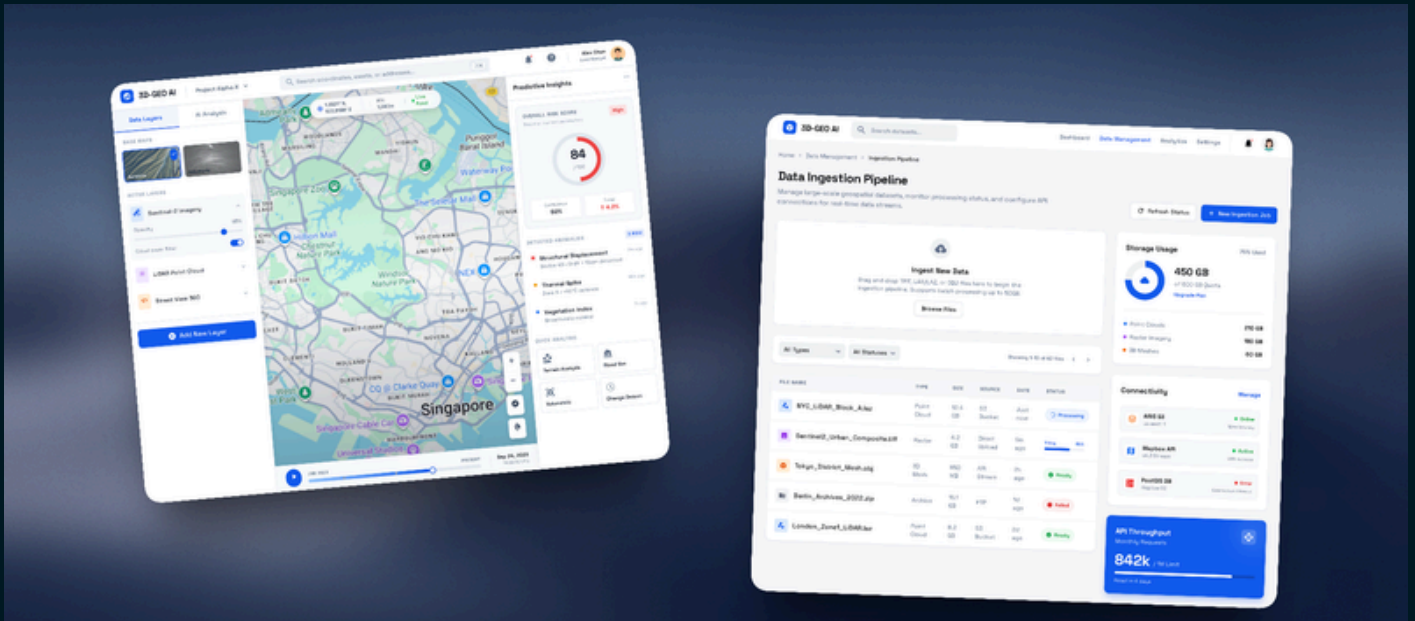
Esri · ArcGIS · QGIS · Mapbox · Cesium · GDAL · PDAL · SamGeo · Leaflet · PostGIS

Web & Mobile

React · Node.js · TypeScript · Swift · Kotlin · React Native

Enterprise 3D Geospatial Intelligence Platform

A SaaS platform to eliminating data silos by integrating satellite imagery, aerial photogrammetry, and ground-sensor feeds into a single, high-performance 3D rendering pipeline for real-time situational awareness.



Industry

Utilities · Public Works · Urban Planning

Role

End-to-end platform design and delivery

Duration & Team

14 months (ongoing) · 8-12 engineers and data scientists

The Challenge

Modern geospatial operations are plagued by fragmented data ecosystems. Analysts must manually cross-reference satellite data, drone point clouds, and vector maps across incompatible software suites. This "swivel-chair" intelligence gathering results in critical latency, with data processing taking up to 48 hours before actionable insights can be derived. Furthermore, legacy rendering engines fail to handle the petabyte-scale datasets required for city-level digital twins.

The Solution

We deployed an end-to-end 3D reconstruction pipeline that ingests satellite and aerial imagery to generate city-scale digital twins automatically. The system uses a proprietary semantic segmentation engine to identify, classify, and vectorize objects in 3D space, integrating directly with existing ERP and GIS workflows.

- **Automated Segmentation:** Classifies terrain, vegetation, and infrastructure with >99% IoU accuracy.
- **Multi-Format Support:** Efficiently handles .LAS, .LAZ, OBJ, & GeoTIFF inputs.
- **Cloud-Native:** Dynamically streams 3D tiles to web clients with zero latency.
- **ERP Integration:** Seamless export to ESRI ArcGIS, Autodesk Civil 3D, and SAP.

TECHNICAL STACK

GEO SPATIAL ENGINE

Cesium PostGis

MACHINE LEARNING

Python Tensor flow

INFRASTRUCTURE

Docker Kubernetes AWS S3

OPERATIONAL IMPACT

45%
Faster Analysis Cycle

Predictive
Modeling Enabled

10GB+
Daily Data Processed

AI-Enabled Early Cancer Detection Platform

A clinical deep-dive into pixel-level diagnostic accuracy & real-time implementation for early-stage pulmonary nodule detection using neural networks.



Industry

Healthcare · Clinical Diagnostics

Role

ML engineering and clinical integration

Duration & Team

12 months · 6-10 specialists

The Challenge

Early-stage oncology diagnostics face a critical bottleneck: the manual review of thousands of high-resolution DICOM slices. Radiologists are under increasing pressure, leading to fatigue-induced oversights in identifying anomalies smaller than 3mm. Current manual screening methods exhibit a baseline accuracy of 82.1%, with a significant 12.5% false positive rate that leads to unnecessary invasive biopsies.

The Solution

We implemented a multi-stage Deep Convolutional Neural Network (DCNN) architecture specifically optimized for pixel-level anomaly detection in volumetric medical data. The system utilizes a custom feature pyramid network (FPN) to maintain spatial awareness across different zoom levels, ensuring that even sub-millimeter nodules are identified with high confidence.

- **Real-time Inference:** Sub-15ms processing per frame allowing for instantaneous radiologist feedback.
- **DICOM Native:** Direct integration with hospital PACS systems without conversion loss.
- **FHIR Compliant:** Seamless electronic health record (EHR) data exchange.

</> TECHNICAL STACK

FRAMEWORKS

Pytorch 2.0 FastAPI CUDA 1.8

DATA STANDARD

DICOM v3.0 HL7 FHIR R4

INFRASTRUCTURE

Python AWS HealthLake Docker

CLINICAL IMPACT

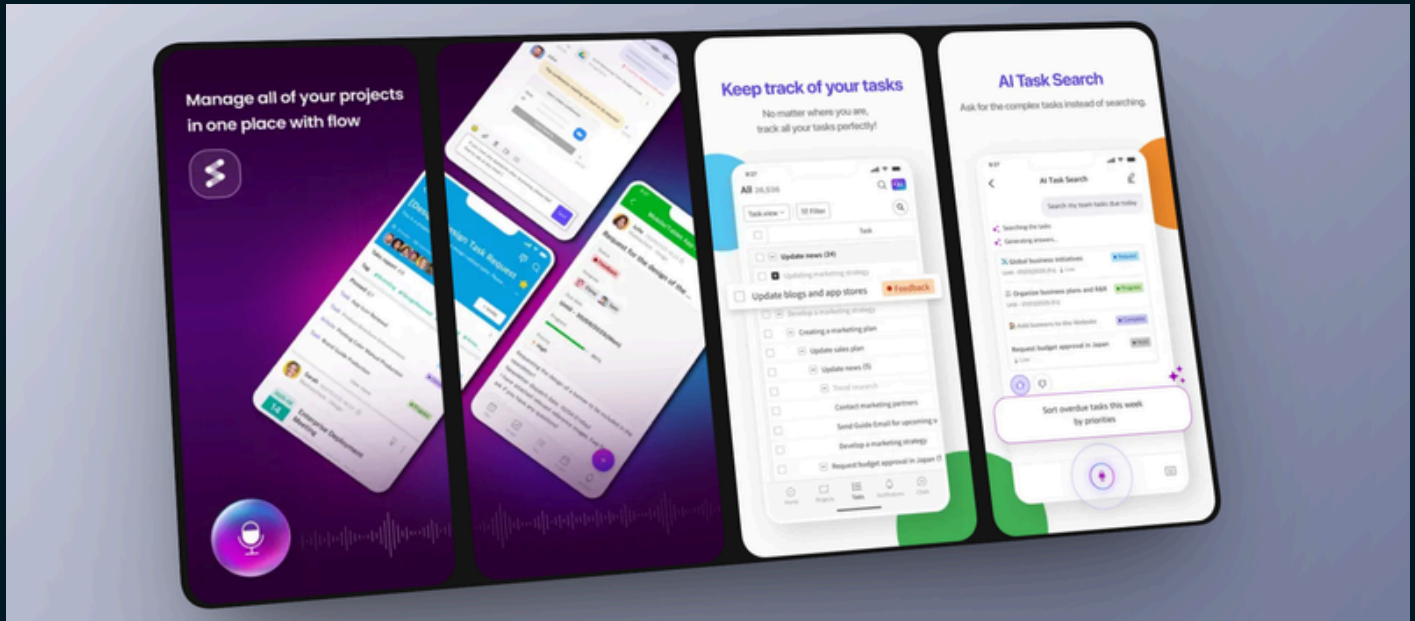
98.4%
Total Detection Rate

2.4x
Efficiency Multiplier

-15%
Operational Cost Reduction

The Operating System for Modern Work

A comprehensive project to streamline organizational workflows by centralizing communication, task management, and file storage into a single, synchronized interface.



Industry

Productivity · Team Collaboration · SaaS

Role

Product delivery and platform support

Duration & Team

18 Months · Cross-functional product and engineering team

The Challenge

Modern distributed teams face significant friction due to disconnected toolchains. Information is fragmented across email, instant messaging apps, and separate task management boards. This "tool fatigue" results in constant context switching, lost data, and a 25% decrease in overall operational efficiency as teams struggle to maintain a single source of truth for their projects.

The Solution

Flow introduces a unified workspace that bridges the gap between communication and execution. By embedding chat directly within task contexts and file repositories, we eliminated silos. The platform leverages a high-performance sync engine to ensure every stakeholder sees the same state, instantly.

- **Task Tracking:** Intelligent assignment and automated status updates based on activity.
- **Collaborative Spaces:** Shared canvases for brainstorming, documentation, and asset management.
- **Unified AI Search:** Find any document, decision, or task across the entire workspace instantly.

</> TECHNICAL STACK

TOOLS

iOS

Swift UI

Python

BACKEND SERVICES

Cloud Backend

RestFul APIs

INFRASTRUCTURE

AWS

Docker

OPERATIONAL IMPACT

2.5 HOURS/DAY

Saved per User on Team

+40%

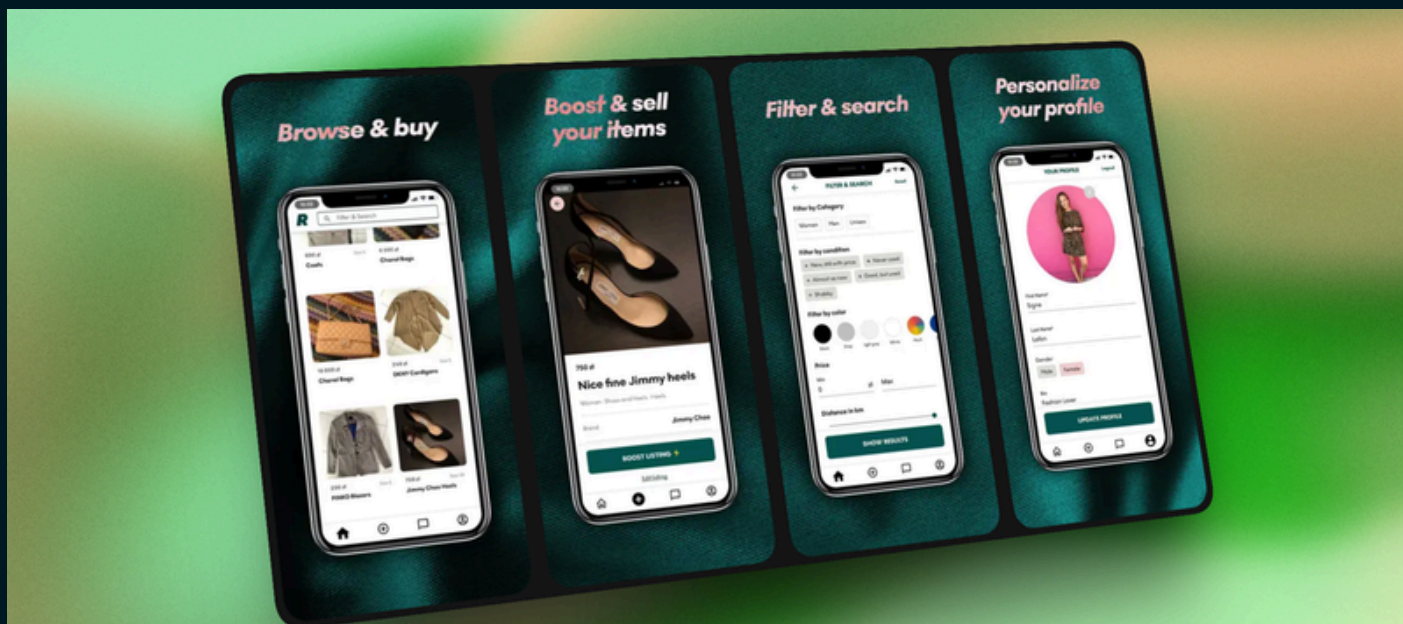
Sprint Velocity

99.9%

Uptime Guaranteed

Ree: The Future of P2P Fashion

A mobile-first marketplace designed to revolutionize the circular fashion economy by building trust, simplifying the listing process, and creating a social shopping experience.



Industry

E-commerce · Fashion · Mobile Marketplaces

Role

Marketplace feature delivery and experience optimization

Duration & Team

Ongoing releases · Mobile and backend development team

The Challenge

Peer-to-peer fashion discovery is traditionally plagued by clutter and lack of trust. Users struggle to find quality items amidst thousands of poor-quality listings, while sellers face high drop-off rates due to complex listing processes. Existing platforms suffer from significant transaction friction and low user retention, making "the hunt" for vintage items more frustrating than rewarding.

The Solution

We implemented a refined item listing workflow that guides sellers through structured data entry, enabling powerful filtering algorithms on the backend. The solution introduces a "Smart Feed" that learns from user interactions to present relevant items, coupled with a secure messaging system that builds trust between buyer and seller before the transaction occurs.

- **Visual Search:** AI-powered image recognition to auto-tag items and reduce listing time.
- **Refined Workflow:** Step-by-step listing wizard reducing seller cognitive load.
- **Escrow Payments:** Funds are held securely until the buyer confirms item condition.

TECHNICAL STACK

TOOLS

iOS

Swift UI

Python

BACKEND SERVICES

Cloud Backend

RestFul APIs

INFRASTRUCTURE

AWS

Docker



OPERATIONAL IMPACT

64%

Sustainable, Repeat Usage

3.2x

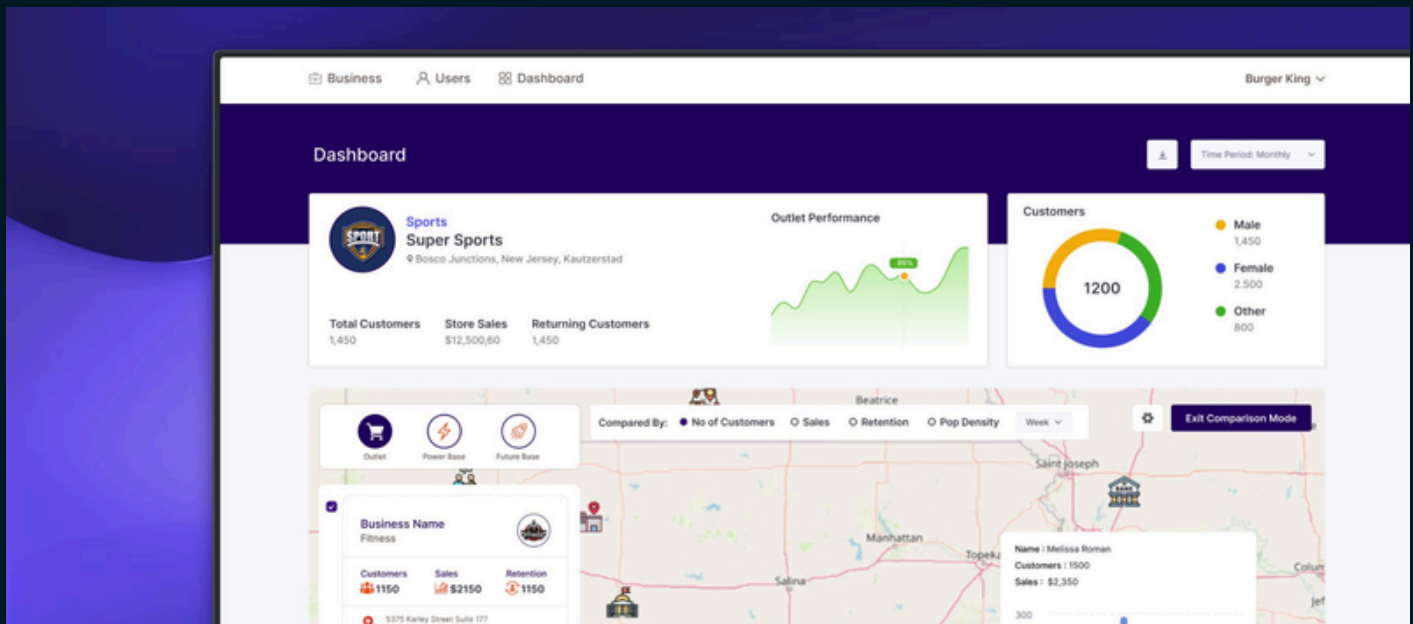
Inventory Turnover

99.9%

Uptime Guaranteed

Business Intelligence & Market Insights Platform

A GIS-Based SaaS platform unifying fragmented business registration data and consumer behavior silos to drive smarter site selection and targeted marketing campaigns.



Industry

Business Intelligence · Retail · Marketing Technology

Role

End-to-end platform design and delivery

Duration & Team

28 months(Ongoing) · 6-10 engineers, data specialists, and product designers

The Challenge

Enterprises and retail chains often struggle with vast amounts of siloed customer data that lacks spatial context. Without the ability to visualize where customers are actually located relative to store locations, businesses face an inability to plan targeted marketing or optimize expansion strategies. This disconnect results in wasted ad spend and missed opportunities in high-potential geographic clusters.

The Solution

We engineered a comprehensive GIS-driven visualization platform that ingests CRM data to generate real-time insights. By utilizing AI-based customer profiling, the system identifies high-value retail clusters. Furthermore, it features integrated Meta and Google ad management, allowing marketers to instantly push geo-fenced campaigns to specific high-performing zones directly from the map interface.

- **GIS-Driven Visualization:** Interactive 3D maps that reveal hidden density patterns.
- **AI Customer Profiling:** Machine learning algorithms that segment audiences by location behavior.
- **Location Recommendation Engine:** Suggests optimal retail sites based on competitor density and consumer footfall.

TECHNICAL STACK

GEO SPATIAL ENGINE

Leaflet PostGis

MACHINE LEARNING

Python Tensor flow

INFRASTRUCTURE

Docker Kubernetes AWS S3

OPERATIONAL IMPACT

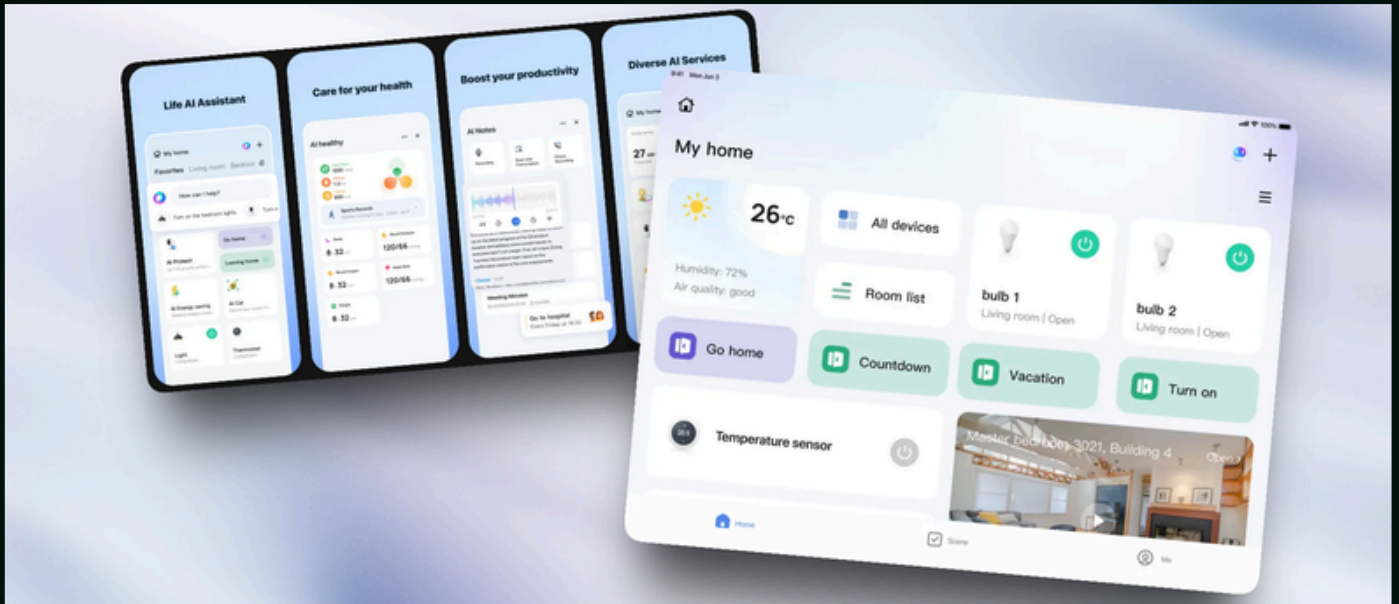
-60%
Reduced Analytics Reliance

92%
Growth Prediction Accuracy

4x
Optimized Marketing

Smart Living & Device Management Platform

A unified digital experience merging smart home controls with personal health metrics, eliminating app fragmentation and enabling context-aware automation.



Industry

Smart Home · IoT · Consumer Application

Role

Mobile platform delivery and ecosystem integration

Duration & Team

Ongoing product development · Mobile, backend, and IoT integration teams

The Challenge

Users needed a single, reliable mobile application to manage and monitor multiple smart home devices while also viewing selected personal health metrics. Existing solutions were fragmented, requiring separate apps for device control and data visibility.

The Solution

Delivered an iOS-based smart living platform that acts as a central hub for controlling connected devices such as lights, sensors, and appliances. Integrated device management with automation and status monitoring to simplify everyday smart home interactions. Enabled HealthKit integration to sync supported body metrics, allowing users to view selected health data alongside their smart living controls.

- **Centralized Hub:** One-tap control for lights, locks, and climate systems.
- **IoT Device Management:** Auto-discovery and grouping of devices by room or function.
- **HealthKit Integration:** Biometric data sync allowing health-triggered home automation.

</> TECHNICAL STACK

TOOLS

iOS SDK

Swift UI

Combine

INTEGRATIONS

Homekit

Healthkit

CLOUDE SERVICES

AWS IoT

Firebase

OPERATIONAL IMPACT

1 APP

Simplified Management

+45%

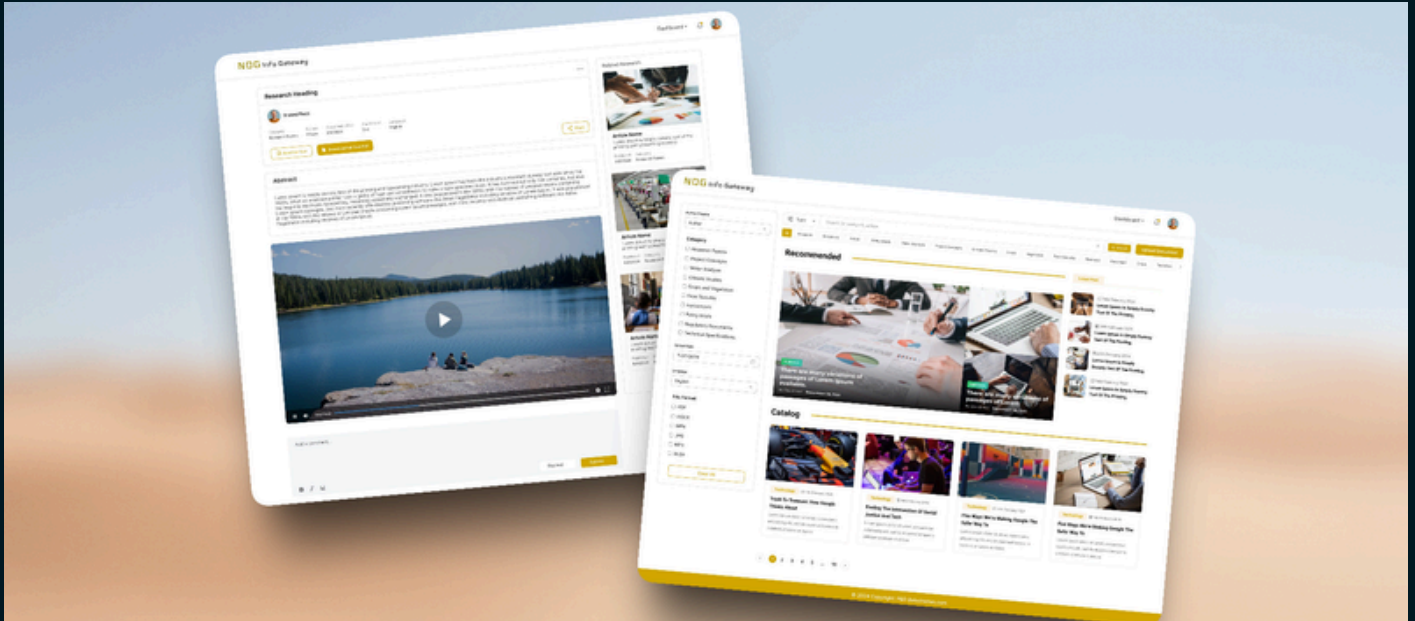
Higher Engagement

Sync

Seamless Health Integration

National Digital Archive & Information Gateway

Unifying 50+ years of archived research, multimedia assets, and institutional records into a single, searchable, governed digital ecosystem.



Industry

Knowledge Management · Public Sector · Research Institutions

Role

Platform design, delivery, and revamp of legacy system

Duration & Team

09 months · 4-6 engineers and product specialists

The Challenge

Government records were deteriorating in obsolete physical formats. Departments operated in isolation with distinct metadata standards, making cross-agency data sharing impossible. The lack of a unified digital strategy meant that critical historical footage and legislative documents were inaccessible to the public and internal researchers alike, creating a "dark archive" of unsearchable assets.

The Solution

We engineered a cloud-native Centralized Portal powered by a robust search engine. The solution introduces a governed publishing workflow that enforces metadata standards upon ingestion. By utilizing automated tagging and optical character recognition (OCR), we transformed static files into searchable, interconnected knowledge assets.

- **Unified Metadata Schema:** Standardized Dublin Core attributes across all media types (Audio, Video, Text).
- **High-Fidelity Video Streaming Engine:** Adaptive bitrate streaming for 4K/8K archival footage with frame-accurate scrubbing and secure delivery.
- **Deep Discovery:** Full-text indexing allows users to search inside PDFs and scanned documents.

</> TECHNICAL STACK

TOOLS

React

Tailwind

TypeScript

BACKEND SERVICES

Cloud Backend

Postgresql

Elastic Search

INFRASTRUCTURE

AWS

Docker

S3 Object Storage

OPERATIONAL OUTCOME

40,000+

Documents Processed

~92%

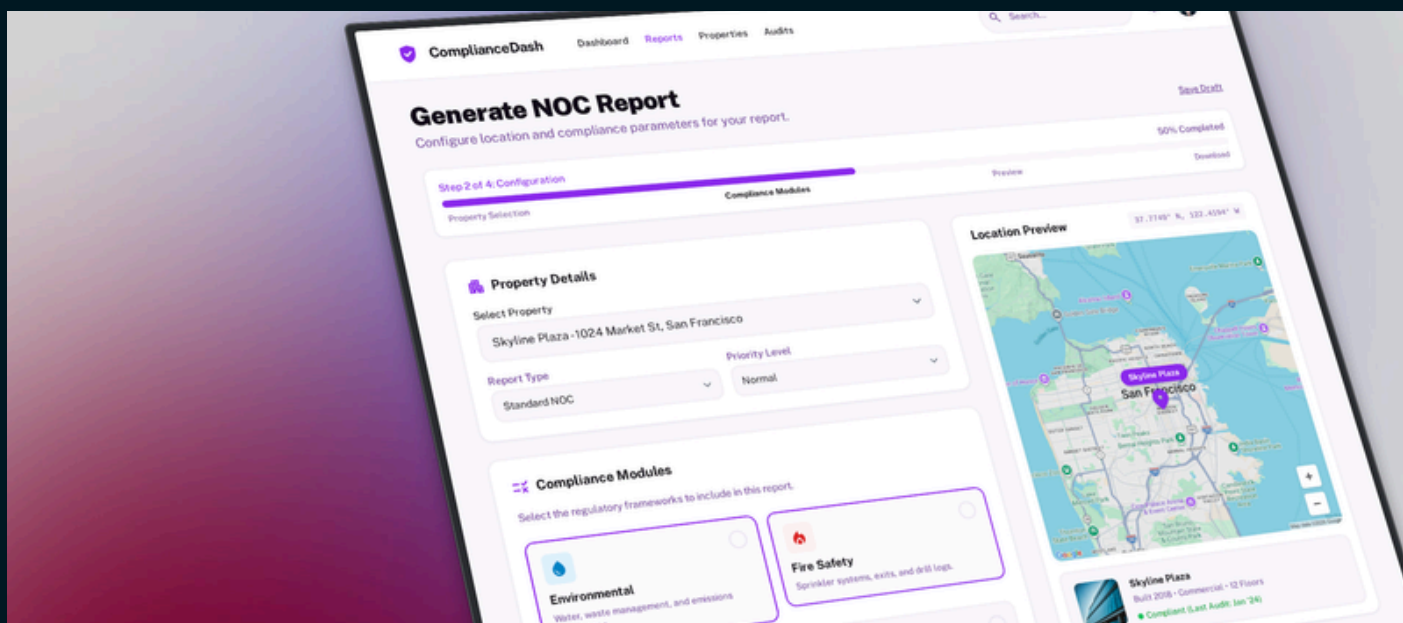
Reduction in Retrieval Time

< 300ms

Query Reterival Time

Property Management & Regulatory Intelligence Platform

We took a chaotic web of local bylaws and turned them into a single, intelligent engine. By using ElasticSearch, we've made it possible for enterprise portfolios to get NOCs automatically with instant data-backed clarity.



Industry

Real Estate · Property Management ·
Public Sector Compliance

Role

End-to-end research, design, delivery
of the platform.

Duration & Team

10 months · 5-7 engineers and domain
specialists

The Challenge

Property managers and stakeholders struggled to manage property records while complying with complex, location-specific government regulations, NOCs, and legislation. Regulatory data was fragmented across departments, making verification, reporting, and due diligence slow and error-prone.

The Solution

Delivered a property management platform that centralizes property records and integrates regulatory and legislative data into a unified system. Implemented ElasticSearch-powered indexing to enable fast retrieval of property details, government approvals, and compliance status. Built configurable reporting workflows that generate detailed, location-aware property reports aligned with local authority rules and NOC requirements.

- **ElasticSearch Retrieval:** Sub-second query performance across 50+ data dimensions (zoning, tax, liens).
- **Smart NOCs:** Algorithmic pre-validation of building plans against local zoning codes.
- **Centralized Repository:** A single source of truth connecting municipal data with developer project files.

</> TECHNICAL STACK

TOOLS

React

Tailwind

TypeScript

BACKEND SERVICES

Elastic Search

Postgresql

INFRASTRUCTURE

AWS

Docker

S3 Object Storage

OPERATIONAL OUTCOME

10X

Regulatory Speed

99.9%

Due Diligence Accuracy

-85%

Compliance Overhead

Meet the Leadership Team



Ahmed Raza

Chief Executive Officer

Provides strategic leadership with a strong foundation in geospatial technologies, guiding the company's vision, growth, and application of AI & spatial intelligence across solutions.



Mr. Osama Shakir

Chief Operating Officer

Oversees global branding, marketing strategy, and sales pipelines while driving operational execution and translating business vision into scalable delivery.



Mr. Humayoun Baig

Chief Technology Officer

Leads technical architecture, engineering workflows, and R&D, ensuring the delivery of AI and GIS-based software systems while maintaining a high-performance engineering culture.



Engagement Models



Fixed Price

Defined deliverables & acceptance criteria for clear project boundaries.



Time and Materials

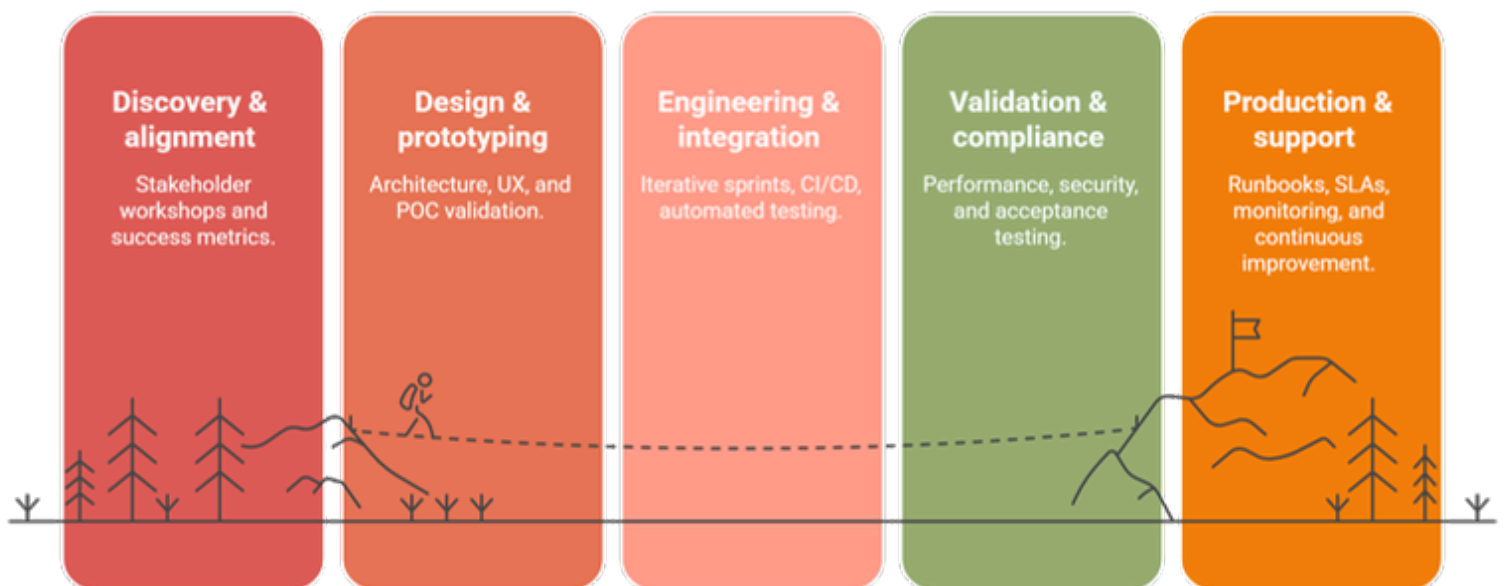
Flexible engagement billed on actual effort and resources used as requirements evolve.



Managed Services

Ongoing operations, monitoring, and continuous feature delivery.

Delivery Lifecycle



Our Clients



Flow



Ree



Smartlife



Victoria Secret



Industries Served

Finance

Healthcare

Retail

Logistics

Education

Government

SERVING WORLDWIDE



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